

Assignment No.4

**Problem Statement : SQL Queries – all types of Join, Sub-Query and View:
Write at least 10 SQL queries for suitable database application using SQL
DML statements.**

Program :

```
SQL> CREATE TABLE Student2 ( student_id INT PRIMARY KEY, name  
VARCHAR(10), age INT, dept VARCHAR(10) );
```

Table created.

```
SQL> CREATE TABLE Enrollment2 ( enroll_id INT PRIMARY KEY, student_id INT,  
course_name VARCHAR(10), marks INT, FOREIGN KEY (student_id) REFERENCES  
Student2(student_id) );
```

Table created.

```
SQL> INSERT INTO Student2 VALUES (1, 'Amit', 20, 'Computer');
```

1 row created.

```
SQL> INSERT INTO Student2 VALUES (2, 'Sneha', 21, 'IT');
```

1 row created.

```
SQL> INSERT INTO Student2 VALUES (3, 'Rahul', 22, 'EXTC');
```

1 row created.

```
SQL> INSERT INTO Student2 VALUES (4, 'Pooja', 23, 'Computer');
```

1 row created.

```
SQL> INSERT INTO Enrollment2 VALUES (101, 1, 'SQL', 85);
```

1 row created.

```
SQL> INSERT INTO Enrollment2 VALUES (103, 2, 'SQL', 92);
```

1 row created.

```
SQL> INSERT INTO Enrollment2 VALUES (104, 3, 'Java', 66);
```

1 row created.

```
SQL> INSERT INTO Enrollment2 VALUES (105, 4, 'Python', 88);
```

1 row created.

```
SQL> select*from Student2;
```

STUDENT_ID NAME AGE DEPT

1 Amit 20 Computer
2 Sneha 21 IT
3 Rahul 22 EXTC
4 Pooja 23 Computer

SQL> select * from enrollment2;

ENROLL_ID STUDENT_ID COURSE_NAM MARKS

101 1 SQL 85
103 2 SQL 92
104 3 Java 66
105 4 Python 88

SQL> --1. **Display all students and their enrolled courses.**

SQL> select enrollment2.course_name, student2.name from student2

2 inner join enrollment2

3 on student2.student_id = enrollment2.student_id;

COURSE_NAM NAME

SQL Amit

SQL Sneha

Java Rahul

Python Pooja

SQL> --2. **List students who have scored more than 80 marks in any course.**

SQL> select name

2 from student2

3 where student_id in (select student_id

4 from enrollment2

5 where marks > 80);

NAME

Amit

Sneha

Pooja

SQL> --3. Find students who are not enrolled in any course.

SQL>INSERT INTO Student2 VALUES (5, 'Shivay', 23, 'EXTC');

1 row created.

INSERT INTO Enrollment2 VALUES (106, 5, ' ', null);

1 row created.

SQL> select s.student_id,s.name

2 from student2 s

3 left join enrollment2 e on s.student_id =e.student_id

4 where e.student_id is null;

STUDENT_ID NAME

5 Shivay

SQL> --4. Display the highest marks obtained in the SQL course.

SQL> select max(marks)as highest_marks

2 from enrollment2

3 where course_name = 'sql';

HIGHEST_MARKS

92

-- 5.highest in python

SQL> select s.name,s.dept,e.marks

2 from student2 s

3 join enrollment2 e on s.student_id =e.student_id

4 where e.course_name = 'python'

5 and e.marks =(select max(marks)

```
6 from enrollment2
7 where course_name ='python');
```

NAME	DEPT	MARKS
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Pooja	computer	88
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SQL> --6 List all departments and average marks scored by their students.

SQL> select s.dept ,avg(e.marks)as avg_marks

```
2 from student2 s
3 join enrollment2 e on s.student_id = e.student_id
4 group by s.dept;
```

DEPT	AVG_MARKS
------	-----------

EXTC	66
------	----

IT	92
----	----

Computer	86.5
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--7. java

SQL> select distinct s.name

```
2 from student2 s
3 join enrollment2 e on s.student_id = e.student_id
4 where e.course_name = 'java';
```

NAME

Rahul

--8 total courses

SQL> select s.name,count(e.course_name) as total_courses

```
2 from student2 s
3 left join enrollment2 e on s.student_id = e.student_id
4 group by s.name;
```

NAME TOTAL_COURSES

shivay	0
Sneha	1
Rahul	1
Amit	1
Pooja	1

--9. max marks of all

```
SQL> select s.name ,max(e.marks) as max_marks
 2  from student2 s
 3  join enrollment2 e on e.student_id =e.student_id
 4  group by s.name;
```

NAME MAX_MARKS

shivay	92
Sneha	92
Rahul	92
Amit	92
Pooja	92

SQL> --10. List students who have scored 92 marks in any course.

```
SQL> select name
 2  from student2
 3  where student_id in(select student_id
 4  from enrollment2
 5  where marks=92);
```

NAME

Sneha